qml by klnsss for msrit Beng 2826 day1 (part2)

for Msrit by sarma

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here the agenda is to express, recap of

- 3 different bases of quantum computing
- CNOT gate(2 qubit), Hadamard gate(1 qubit),Rotation gates (1 qubit)
- Phase kick
- Deutsch algo
- QPE (quantum phase estimation, also QFT)
- Relevant number theory results for Shor's Algo.
- Grover's Algo.
- Quantum speedup

3 different orthonormal bases of quantum computing

- Z-basis, X-basis (Hadamard basis / diagonal basis), Y-basis (circular basis)
- Z-basis $\mapsto$ called Energy basis (correspond to qubit energy levels)
- $\mapsto$ computational basis / standard basis / Z - basis states
- $|0\rangle \mapsto$ top of Bloch sphere
- $|1\rangle \mapsto$ opposite to $|0\rangle$
- $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Hadamard basis / X-basis

- state vectors are
- $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- associated observable is —pauli X matrix

  $\mathrm{X} = |+\rangle\langle+| - |-\rangle\langle-| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Circular basis / Y-basis

- state vectors are
- $|+_i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix}$
- $|-_i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$
- associated observable is —pauli Y matrix

  $\mathrm{Y} = |+_i\rangle\langle+_i|\,(-)\,|-_i\rangle\langle-_i| = \begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}$

Basis transformation operator 'U'

- $|U\rangle = |+\rangle\langle 0|(+)|-\rangle\langle 1|$
- $\Downarrow$
- $|+\rangle = U|0\rangle \implies |+\rangle\langle 0| = U|0\rangle\langle 0|$
- $|-\rangle = U|1\rangle \implies |-\rangle\langle 1| = U|1\rangle\langle 1|$

Hadamard Gate

- used to create superposition, since, it $\xrightarrow{transforms} |0\rangle \mapsto |+\rangle$, and $|1\rangle \mapsto |-\rangle$
- $H^2 = I(involutory)$
- $H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- $H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- $H|+\rangle = |0\rangle$
- $H|-\rangle = |1\rangle$

parametric quantum gates

- class of gates which depend on a continuous parametre, usualy an angle of rotation—parametric quantum gates
- these gates ensure— control over qubit states (extensively used in quantum Algorithms)
- Rotation gates $\Leftrightarrow$ allow transformations around Bloch's sphere's axes
- Universal gate sets enable any quantum operation of need.
- for any arbitrary single-qubit operation,rotation gates are used to manipulate qubit states.

R_x, R_y, R_z matrices

- $R_x(\theta)\,|\psi\rangle = \cos\frac{\theta}{2}\,|\psi\rangle - i\sin\frac{\theta}{2}X\,|\psi\rangle \mapsto$ when applied
- $R_x(|\theta\rangle) = [\cos\frac{\theta}{2} - i\sin\frac{\theta}{2} - i\sin\frac{\theta}{2}\cos\frac{\theta}{2}] \mapsto$ matrix
- $R_y(|\theta\rangle) = [\cos\frac{\theta}{2} - \sin\frac{\theta}{2} - \sin\frac{\theta}{2}\cos\frac{\theta}{2}] \mapsto$ matrix
- $R_y(\theta)\,|\psi\rangle = \cos\frac{\theta}{2}\,|\psi\rangle - \sin\frac{\theta}{2}Y\,|\psi\rangle \mapsto$ when applied
- $R_z(\theta) = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$
- $R_z(\theta)\,|\psi\rangle = e^{-\frac{\theta}{2}}\,|0\rangle + e^{-\frac{\theta}{2}}\,|1\rangle \mapsto$ when applied to a qubit state
- $R_x \mapsto$ Rotation around x-axis
- $R_y \mapsto$ Rotation around y-axis
- $R_z \mapsto$ Rotation around z-axis

Minimal universal gate sets

- UNiversal gates: a small collection of gates capable of approximating any quantum operation to arbitrary precision and are fundamental for quantum computing.(look at next slide for tabular info)
- clifford gates: (H,S,CNOT),
- T gate$\mapsto$ non clifford but essential for. || phase shift gate with $\phi = \frac{\pi}{4}$ universality
- T gate matrix $= \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{bmatrix}$
- useful in grover's and or Shor's factoring algorithm

Universal gate sets

universal gate set	elaboration
{H,T,CNOT}	minimal clifford+T set
{H,S, T,CNOT}	extended clifford +T
$\{R_x(\theta), R_y(\theta), CNOT\}$	parametrized for single qubit
{X,Y,Z,H,S,T,CNOT}	for simulation rep'n
Hardware basis gates	native to devices(ignore)

Phase kickback

- WHEN CX (CNOT) gate is applied to the target value $(\frac{|0\rangle - |1\rangle}{\sqrt{2}})$ with control qubit $|1\rangle \implies$
- $CX\,|1\rangle\,.\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = |1\rangle\,.\left((-1).\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\right) \implies$
- $CX\,|1\rangle\,.\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = -\,|1\rangle\,.\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \implies$
- nothing happens to control qubit $|0\rangle$, no change, $\implies$
- $CX\,|0\rangle\,.\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = |0\rangle\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \Leftrightarrow$ phase(target value) is kicked back to register
- CNOT (2 qubit) matrix is in next slide.

CNOT gate matrix / details

- $$\mathrm{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & X \end{bmatrix}$$
- $$\begin{array}{c} \\ |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{array} \begin{array}{c} \begin{array}{cccc} \langle 00| & \langle 01| & \langle 10| & \langle 11| \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

CNOT operator

- $CNOT\,|\psi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|11\rangle + \delta\,|10\rangle$
- $CNOT\,|{+}{+}\rangle = |{+}{+}\rangle$
- $CNOT\,|{+}{-}\rangle = |{-}{-}\rangle$
- $CNOT\,|{-}{+}\rangle = |{-}{+}\rangle$
- $CNOT\,|{-}{-}\rangle = |{+}{-}\rangle$

Quantum Algorithms (QML aligned) intro

Agenda here is :

- concept of quantum speedup
- Deutsch-Jozsa algorithm (more complex and appropriate for QML than simpler 'Deutsch'(prescribed))
- Grover's algorithm
- Shor's Algorithm

Quantum speedup

- quantum speedup—potential computational benefit / advantage that quantum algorithms have over classical algorithms in solving certain types of computational problems.
- 2 types of quantum speedup
- 1)—POLYNOMIAL SPEEDUP
- 2) —— exponential speedup
- polynomial speedup—- if classical computation time=n^k,and. quantum computation time will be—- n^b where b<k
- exponential speedup—- exponential time for classical computation where as , only , polynomial time on a quantum computer.

time complexity notation of Algorithms

- T(n)—classical time complexity of size 'n'
- Q(n)——quantum time complexity for same 'n'
- quantum speedup S(n) = $\frac{T(n)}{Q(n)}$
- larger the s(n)—-greater speedup
- S(n)—-is exponential $\implies$ quantum algorithm is exponentially faster as compared to best known classical algorithm

examples of quantum speedup

- shor's algorithm for integer factorization
- shor's classical factoring— exp $\left(\mathcal{O}((logN)^{\frac{1}{3}}(loglog\mathcal{N}^{\frac{2}{3}}\right)$
- shor's quantum version—$\mathcal{O}((logN)^2(loglogN)(logloglogN)$ is polynomially faster.
- Grover's Algorithm $\xrightarrow{quadraticspeedup}$ for unstructured search problems
- Grover's classical——$\mathcal{O}(\mathcal{N})$ $---$ *lineartimetofindtarget*
- Grover's quantum—— $\mathcal{O}(\sqrt{\mathcal{N}})$—-quadratic improvement

problem representation of quantum algorithms

- quantum algorithms reformulate classical problems into unitary operations compatible with quantum systems.
- popular QML examples– (i) optimization problems (ii) variational quantum algorithms
- quantum algorithms exploit the properties like (a) superposition (b) entanglement, (c) interference to solve computational problems more efficiently than classical algorithms in specific contexts
- quantum algorithms process information in parallel by manipulating quantum states in superposition
- quantum capability provides speedups / exponential speedups

Deutsch-Jozsa Algorithm, and few other qml problems intro

- enables quantum speedup for Decision problems
- Def'n of decision problems—problems for which certain/ some algorithms exist which always halts with the correct 'yes' or 'no' answer
- Decision problems are also called Turing-decidable / computable problems.
- VQE,QAOA solve optimization challenges.
- QNN's, QSVMs integrate quantum power into classical models (enhanced / inspired)

Deutsch-Jozsa core ideas

- determines whether a function f(x) is constant or balanced using quantum parallelism.
- f(x) is constant if it outputs the same value for all inputs
- Balanced if it outputs an equal number of $|0\rangle, |1\rangle$ solutions.
- for a function of 'n' bits of input $\Longrightarrow$
- with classical evaluation $\xrightarrow{requires} 2^{n-1} + 1$ queries (worst case)
- with quantum algorithm—- solved with just 1 query.

balanced / quantum function details–tabular

x	$f^{00}(x)$	$f^{01}(x)$	$f^{10}(x)$	$f^{11}(x)$
0	0	0	1	1
1	0	1	0	1
	Constant	Balanced	Balanced	Constant

Figure: from a publication with thanks

simple case– direct Deutsch algo relevance

XOR function

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

Figure: from a published book with thanks

another table (XOR) in D-J context

Property	Name
$x \oplus x = 0$	X1
$x \oplus 0 = x$	X2
$(x \oplus y) \oplus z = x \oplus (y \oplus z)$	X3

Figure: taken from a published book with thanks

4 oracles of Deutsch-Jozsa with $8(2^3)$*possibilities*

x	$f_0(x)$	$f_1(x)$	$f_2(x)$	$f_3(x)$
000	0	1	0	1
001	0	1	0	1
010	0	1	0	0
011	0	1	0	1
100	0	1	1	0
101	0	1	1	0
110	0	1	1	0
111	0	1	1	1

Figure: with thanks to the publisher / author

outlines of D-J implementation

- circuit initialization (step 1)
- starts with 2 qubits
- 1—in the superposition state for input
- $|\psi\rangle = H^{\otimes n} |\{0\rangle^{\otimes n} \bigotimes H |1\rangle$
- another (2nd qubit) as auxiliary qubit for the oracle
- oracle Application (2nd step)— to encode the function f(x)
- Result measurement (3rd step)— Hadamard gate applied again and output qubits are measured

Variational quantum Algorithms

- VQA's use hybrid quantum-classical approaches to solve problems like—— optimization and eigen spectrum determination
- VQE (variational quantum eigensolver) is a kind / type of VQA—that minimizes the expected energy of a molecule's Hamiltonian H using parametrized quantum circuits
- optimization problems mapped into a quantum Hamiltonian Operator H
- H=$\sum_{i,j} c_{ij}\sigma_i^2\sigma_j^2 + \sum_i b_i\sigma_i^2$
- $\sigma_i^2 \mapsto$ pauli Z operator acting on the ith qubit
- c_{ij}, b_j are problem specific coefficients
- $E(\theta)_{VQE} = \langle\psi(\theta)|H|\psi(\theta)\rangle$
- classical optimizer adjusts parameters θ to minimize $E(\theta)$

Quantum approximate optimization Algorithm(QAOA)

- QAOA is a combinatorial optimization problem
- by iteratively applying the cost Hamiltonian
- In machine learning (classical)— `cost function (error function)` is for the entire data set, unlike loss function which for a data point.
- cost function helps to measure the extent to which the model is going wrong in estimating the relationship between $\mathcal{X}, \mathcal{Y}$ kind of R^2 function in regression model
- Machine learning itself is an optimization problem —-using 'objective function'
- Hypothesis is measured using a 'cost function' in machine Learning
- in NN / QNN (quantum neural nets), recall— gradient descents, SGD(stochastic gradient descents) are optimization techniques only

Elementary note on QAOA

- QAOA solves Combinatorial optimization problems by encoding the cost function H_C into a Hamiltonian and using a mixing Hamiltonian H_M to explore solution space
- $|\psi\rangle = e^{-i\beta H} e^{-i\gamma H_C} |\psi_0\rangle$
- $|\psi_0\rangle - - -$ *initialstate*,β, γ are tunable parameters
- QAOA popular applications are `scheduling, route optimization,resource allocation in engineering`

cost Hamiltonian H_C of QAOA

- $\mathcal{H}_\rfloor = \sum_{i<j} J_{ij} Z_i Z_j + \sum_i h_i Z_i$
- 2 components— (i)Applying cost Hamiltonian (ii) Mixing operator
- quantum circuit alternates between the above 2 components.
- optimization components of QAOA
- GOAL is taken care of by objective function
- input data $\leftrightarrow$ decision variables
- constraints $\leftrightarrow$ limitations on the variables

problem definition of Grover's Algorithm

- given a function 'f' that takes 'n' bits as input and yields a single bit as output, mathematically $\mapsto f : 0, 1^n \longmapsto 0, 1$
- usecase : search problems—- to find some specific sequence of bit as output, within a vast collection of other sequences. i.e a function checks each sequence and returns True(1) if it matches X i.e f(x)=1
- the oracle and diffusion steps are approximately $\frac{\pi}{4}\sqrt{2^n}$ item diffusion operator amplifies the probability of the correct state.

single Mathematical basis of Grover's Algorithm

- the state evolution in Grover's algorithm amplifies the amplitude of the desired state.
- after $\sqrt{N}$ iterations, probability of measuring the target state approaches 1
- $|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle \mapsto |x_i\rangle$
- $|x_i\rangle$ is the marked one.
- in between transformation $\mathcal{U}_w$ is applied to the state which flips the sign of the marked item
- orcle query in grover's $\mathcal{O}|x\rangle = \begin{cases} -|x\rangle - - - x = x_0 \\ |x\rangle \; else \end{cases}$

in between transformation $\mathcal{U}_w$ of grover's and Diffusion operator

- $\mathcal{U}_w = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} y_x \left|x\right\rangle$
- post 'oracle application' another transformation= grover's diffusion operator (Grover iterate($\mathcal{U}_s$)) is applied—— to perform an inversion about the average amplitude
- Grover's iterate $\mathcal{U}_s = 2\left|s\right\rangle\left\langle s\right| - I$
- where $\left|s\right\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \left|x\right\rangle \mapsto$ equal superposition state and I is the identity operator.
- next is iteration stage.

Grover's algorithm in Brief

- speedup facts have already been discussed in this day1 part 2 only, so not repeating again.
- Grover's algorithm initializes the system in an equal superposition of all states
- $|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |x\rangle$
- it then iteratively applies 2 operations
- (i) Oracle query—-flips the phase of the marked element. 2 stages of grover' oracle are— (a) G_1–marking (b) G_2 diffusion
- Grover's oracle is iterated.
- Amplitude amplification—enhances the probability amplitude of the marked state through a reflection operation.

Shor's algorithm

- shor's algo. uses QFT (quantum Fourier transform) to find the periodicity of a function.,
- using $f(x) = a^x mod N$
- where N–number to be factorized, a—-randomly chosen integer coprime to N
- QFT $\mapsto$ quantum state from time domain to the frequency domain, identifying period r
- QFT mapping $|x\rangle \mapsto \frac{1}{\sqrt{r}} \sum_{i=o}^{r-1} e^{\frac{2\pi ikr}{r}} |k\rangle$
- once 'r' is determined, N can be factorized by computing $gcd(a^{\frac{r}{2}} \pm 1, N)$

bird's eye view of terms / concepts used in shor's algorithm

strong advice : all this background is number theory of mathematic—- can be ignored in first reading / exam purpose

- order—order of element 1 in $\mathbb{Z}_m$ is equal to 'm', as taking a sum of 'm' times= identity element.
- Example —consider $\mathbb{Z}_{10}$, here 2 has order '5 since 2+2+2+2+2=0 mod 10'. order of 3=10 as the smallest multiple of 3 which has remainder 0 mod 10 is 30= 10×3 = 3 +3+3+3+3+3+3+3+3+3+3+3
- General formula is
- $\frac{LCM(m,k)}{k} = \frac{m}{GCD(m,k)}$
- Multiplicative inverse:Remainders a , b in $\mathbb{Z}_m$ are called multiplicative inverses of each other if ab=1 mod m

contd

- in $\mathbb{Z}_{10}$ elements 3 and 7 are inverses of each other, as 3X7=1 mod 10, 2 is not invertible as there no remainder b such that 2b=1 mod 10 $\mapsto$ basis of Euclid's algorithm

period finding example for shor's Algorithm

- for a sequence 's' created from a list of r=4 numbers repeated m=3 times for a list of 12 elements.
- sequence s= $\underbrace{3,7,2,4}, \underbrace{3.7,2,4}, \underbrace{3,7,2,4}$
- 'r' is the repeated length period of the sequence that nakes up the list, 'm'–number of repeats of that list in 's'
- considering 'r' divides 'N' case i.e $m = \frac{N}{r}$, m –integer.
- considering a function f(x)=x mod r, then for inputs x=0 to r-1 $\implies$ the outputs of f(x) from 0 to r-1 and...

value of m mod 4 for increasing value of m

inputs x=0 to r-1 x=r-1 to produce outputsof f(x)=0 to r-1 ,
inputs x=r to x= r to 2r-1 and again

m	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
m mod 4	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2

sequence of 4 elements being repeated 3 times

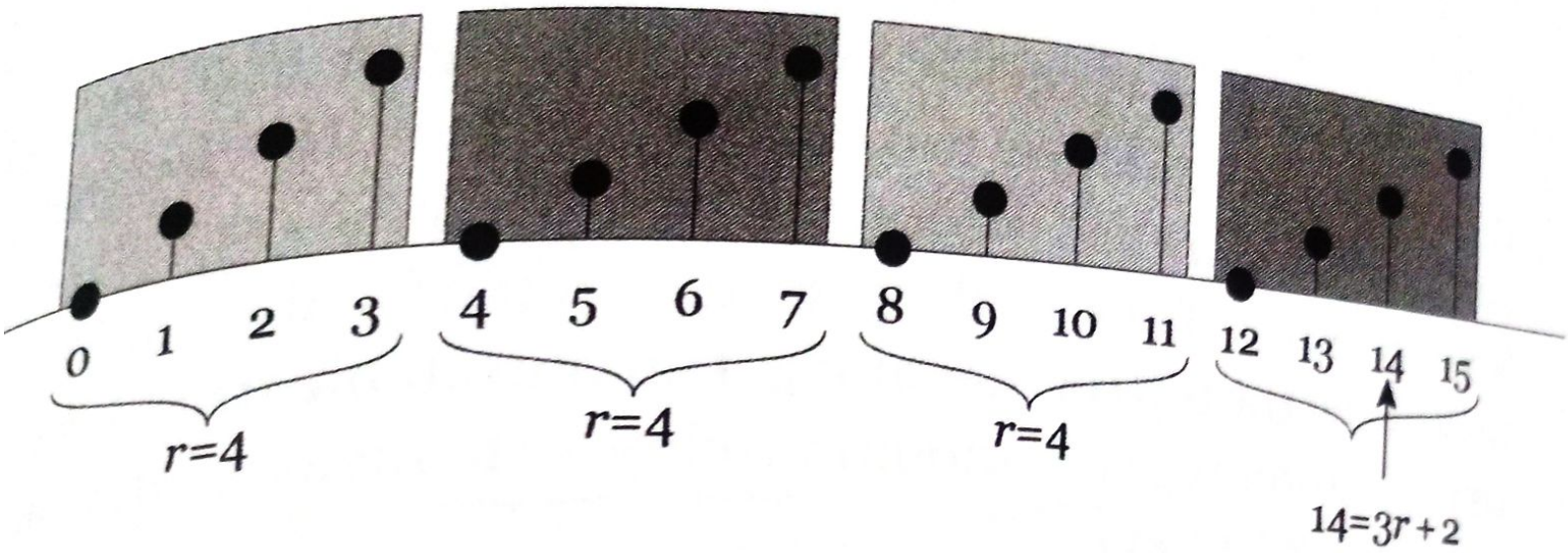

Figure: from a printed book

shor's Illustration

x	7^x	$7^x \bmod 15$
0	1	1
1	7	7
2	49	4
3	343	13
4	2,401	1
5	16,807	7
6	117,649	4
7	823,543	13
8	5,764,801	1
9	40,353,607	7
10	282,475,249	4
11	1,977,326,743	13

Figure: from a published book

shor's algo initial steps

1. **Setup:** We're given $M = 15$.
2. **Pick circuit size:** Because $M^2 = 225$, any value of $n \geq 8$ will do, as $2^8 = 256 > 225$. Let's arbitrarily pick $n = 11$.
3. **Pick a:** Arbitrarily pick $a = 7$.
4. **See if we're lucky:** Find the gcd of a and M. Because $\gcd(7, 15) = 1$, we weren't lucky, so we continue.
5. **Go quantum:** Build the quantum gate that implements the function

Figure: from a published book with thanks

shor's algorithm in 5 steps

- take a value and raise to the power r=r^x and find how bigger it is, compared to m.N and see $r^x - m.N >$. quantum computer puts r^x in a superposition state in which x=1,2,3.... at same time
- 2nd stage——QC uses superposition to calculate 'x',$r^x - m.N >$ where x= 1,2,3... (benefit of quantum computer)
- if $r^x = m.N + 1$, then $\exists$ a condition where $r^{x+p} = m_1, N + 1$. these values maintain a frequency. such frequency can be calculated by inversing period, where `QFT` is useful

contd, from 4th step of shor's

- `QFT` can perform destructive interference and can give one value of 'p'. if p is even,then substitute it in $r^{\frac{p}{2}} \pm 1$ to get the factors.
- if p is odd then start again from guess (random number)

worked out example—N=15 factorization (shor's)

on a quantum computer , procedure of factorization :

1. take an integer x that is less than and prime to N, say x=7
2. to find order 'r' of the congruence $x^r \equiv 1 mod n$, perform the Hadamard transformation 'n' times on the first register of the initial state $|0\rangle^{\otimes n} |1\rangle = \frac{1}{\sqrt{n}} \sum_{a=0}^{2^n-1} |a\rangle |1\rangle$, with n=11,
3. find f(a)$\equiv x^a mod N$ and put the result in 2nd register i.e $\frac{1}{\sqrt{2^n}} \sum_{a=0}^{2^n-1} |a\rangle |x^a mod N\rangle = \frac{1}{2^n} \Big(|0\rangle |1\rangle + |1\rangle |7\rangle + |2\rangle |4\rangle + |3\rangle |13\rangle + |4\rangle |1\rangle + |5\rangle |7\rangle + |6\rangle |4\rangle + ...$
4. measure the 2nd register to find r=0,4,8,... satisfying $x^r \equiv 1 mod N$

shor's factorization example step 5(contd)

step 5:
from $x^{\frac{r}{2}} \pm 1 = 7^2 \pm 1 = 48, 50$
gcd(48,15)=3 and gcd (50,15)=5
$\implies$ 3 and 5 as the prime factors of N=15.

QFT basic info as appendix to Shor's algo

DFT$\leftrightarrow$ *QFT* analogy

- DFT (discrete Fourier transform) in classical signal processing$\equiv$ QFT(quantum Fourier transform) in quantum computing.
- both DFT, QFT Time domain to $\longmapsto$ frequency domain transformation.
- starting from DFT,i.e transformation from one vector to another
- $[x_0, x_1, ..x_{N-1}] \overset{[}{\underset{dft}{\longleftrightarrow}} idft] bothways\ [f_0, f_1, d_{N-1}]$
- after normalization $\Longrightarrow$
- $f_k = \left(\frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{\frac{-i2\pi k_j}{N}} \right.$
- putting $e^{\frac{i.2\pi}{N}} = \omega \Longrightarrow$

QFT contd

- $f_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega^{-kj}$
- above equation = kind of `Matrix times a vector`
- so, operator must be `unitary`
- so, `QFT with 1 qubit=Hadamard only` (homework)
-